

Multiple Choice (10 marks)

1. Those being hunted adopt a coloring scheme that allows them to blend in to the colors of the environment.
 - (a) Mimetic coloration
 - (b) Aposomatic coloration
 - (c) Deceptive markings
 - (d) Batesian mimicry
 - (e) Flash coloration
2. What is meant by the term tonus, or tone?
 - (a) Tonus refers to the electrical activity in the brain during relaxation.
 - (b) Tonus refers to the maximum force a muscle can exert in a single effort.
 - (c) Tonus refers to the flexibility and range of motion in joints and muscles.
 - (d) Tonus refers to the state of complete relaxation in skeletal muscles.
 - (e) Tonus refers to the involuntary contraction of smooth muscles in the digestive tract.
3. During the courtship of the common tern, the male presents a fish to another tern. Account for this behavior and discuss why courtship in general may be necessary for reproduction.
 - (a) Courtship is primarily for entertainment purposes among terns
 - (b) Courtship is not necessary for reproduction
 - (c) Courtship is a way to show dominance
 - (d) Courtship is a random behavior with no impact on reproduction
 - (e) Courtship is necessary for physical transformation required for reproduction
4. Why would you not expect conjugation among a group of paramecia that had descended from a single individual through repeated fission?
 - (a) conjugation results in no genetic diversity
 - (b) conjugation requires more than two organisms
 - (c) all individuals are identical genetically, and conjugation could not occur
 - (d) repeated fission in paramecia results in sterile offspring that cannot conjugate
 - (e) the group of paramecia lacks the cellular structures necessary for conjugation
5. The Krebs cycle in humans occurs in the

- (a) peroxisome
- (b) cytoplasm
- (c) nucleus
- (d) intermembrane space
- (e) outer mitochondrial membrane

True / False (10 marks)

Answer True or False

- 6. True or False: Lysosomes and endosomes can be effectively separated by low-speed centrifugation because they have similar sizes and densities.
- 7. True or False: The Hardy-Weinberg law requires that genes are not exchanged with other populations for equilibrium to be maintained.
- 8. True or False: Meiosis in this organism would produce cells with 32 chromosomes.
- 9. True or False: Crossover is more likely to occur between genes located far apart on the same chromosome than between closely linked genes.
- 10. True or False: Cyclic electron flow through photosystem I results in the synthesis of water from oxygen and protons.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the mechanisms by which certain organisms maintain a constant body temperature despite fluctuations in environmental temperature, highlighting the physiological adaptations involved.
- 12. Discuss the role of bacteria in the dairy industry, focusing on their contributions to the production of fermented milk products and the nutritional processes in dairy animals.

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